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Day 3 :- Review report on three research paper reference

Reference 1 :- Vision Transformer-Based Tea Leaf Disease Classification Using a Novel Dataset

Introduction :- Tea is one of the most important cash crops, contributing significantly to the national economy. However, its cultivation is frequently threatened by various leaf diseases that affect yield and quality. In this study, it introduces a novel, expert-annotated image dataset consisting of 3,300 tea leaf images—spanning six major diseases and healthy samples—collected from key tea-producing regions of Bangladesh under natural field conditions. The dataset is designed to facilitate the development and evaluation of automated disease detection systems using deep learning techniques. Each image has been meticulously labeled by tea experts based on field diagnosis and visual symptoms, ensuring biological and agricultural validity. This work addresses domain-specific challenges such as class imbalance, inter-disease visual similarity, and noisy backgrounds. The dataset is partitioned into training, validation, and test sets (80:10:10) to ensure effective model development. This dataset serves as a foundational resource for researchers aiming to build scalable, AI-driven plant health monitoring systems tailored to the tea industry in developing regions.

Dataset used :-

Challenges in collecting the dataset :- New developments in computer vision and deep learning, especially in Convolutional Neural Networks (CNNs), have recently enabled new directions in automatic detection of plant diseases based on image-based methods. A reliable and diverse dataset is essential for training such models, especially for tasks requiring fine-grained classification under variable natural conditions. While existing datasets like PlantVillage have demonstrated success in controlled environments, there remains a scarcity of publicly available datasets that reflect real-world complexities, especially for region-specific crops like tea

Dataset :- In this thesis, we present a comprehensive image dataset of tea leaves, captured from multiple tea gardens across Sylhet, Panchagarh, and Moulvibazar—three major tea-producing regions of Bangladesh. The dataset includes 3,300 images covering six common disease categories—Anthracnose, Algal Leaf, Bird Eye Spot, Brown Blight, Red Leaf Spot, and White Spot—alongside a healthy leaf category for control comparison. All samples were captured in natural lighting conditions using mobile cameras, with expert-verified annotations based on observable symptoms and field assessments.

The dataset was split into training, validation, and test sets with a ratio of 80:10:10 to ensure a balanced distribution across all categories.

Challenges associated with this dataset include:

- Visual similarity between some disease types (e.g., Red Leaf Spot and Bird Eye Spot).
- Background noise and variable lighting in field conditions.
- Minor class imbalance, particularly in the Brown Blight category.



(a) Healthy leaf



(b) Tea leaf blight



(c) Tea aglae leaf spot



(d) Tea red scab



(e) Tea bud blight



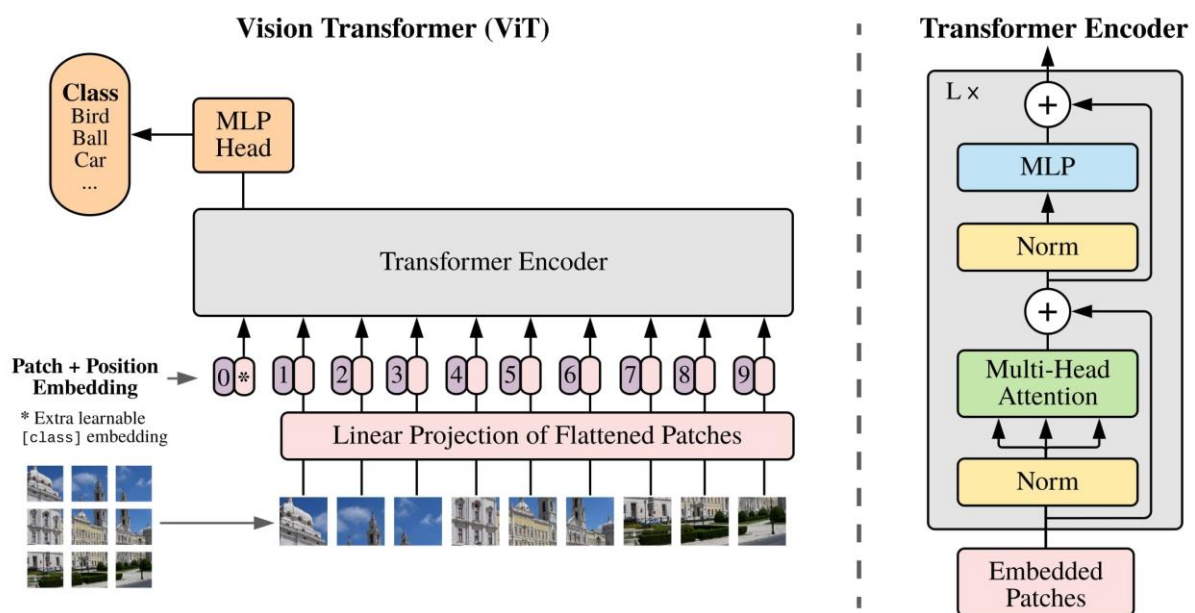
(f) Tea grey blight

Methods Used :- At first various machine learning classifier models such as Support Vector Machine, KNN , Random Forests were used but the problem is they failed to capture the complex disease patterns in the tea leaf.

So CNN based pretrained models such as AlexNet, VGG, ResNet and various inception models were used . Although they are able to capture and extract the complex features from the images and capture the disease patterns , but their accuracy were not so good for smaller datasets as they overfit them.

Vision Transformer Based Approaches :-

Vision Transformers (ViTs) have recently emerged as a powerful alternative to CNNs, leveraging self-attention mechanisms to capture global dependencies in images . In agriculture, ViTs have shown promise for tasks requiring finegrained pattern recognition. applied ViTs to rice leaf disease classification, outperforming CNNs in certain scenarios. Similarly, used ViTs for tomato leaf diseases, highlighting their ability to model long-range dependencies. Hybrid models combining CNNs and transformers, have also been explored to balance local and global feature extraction. For tea leaves, employed a ViT variant for disease detection, though limited to a smaller dataset. Studies note ViTs' advantages in handling diverse image conditions but emphasize their high computational cost and data requirements . Recent advancements, such as efficient ViTs and lightweight transformers , aim to address these limitations.



Results :-

Class	Precision	Recall	F1-Score	Support
Anthracnose	0.96	0.94	0.95	80
Algal Leaf	0.95	0.96	0.95	60
Bird Eye Spot	0.94	0.95	0.94	70
Brown Blight	0.97	0.96	0.96	70
Healthy	1.00	1.00	1.00	50
Red Leaf Spot	0.98	0.99	0.98	90
White Spot	0.97	0.96	0.96	80
Accuracy			0.97	500
Macro Avg	0.97	0.97	0.97	500
Weighted Avg	0.97	0.97	0.97	500

Remarks :- While the proposed Vision Transformer model achieved a high accuracy of 97% on the Tea Sickness Dataset , there remain opportunities for improvement. The confusion matrix revealed minor misclassifications between visually similar diseases, such as Anthracnose and Brown Blight, due to overlapping symptoms like leaf discoloration. To address this advanced data augmentation techniques to enhance the model's ability to differentiate subtle patterns, as well as integrate hybrid architectures combining CNNs and transformers for improved feature extraction.

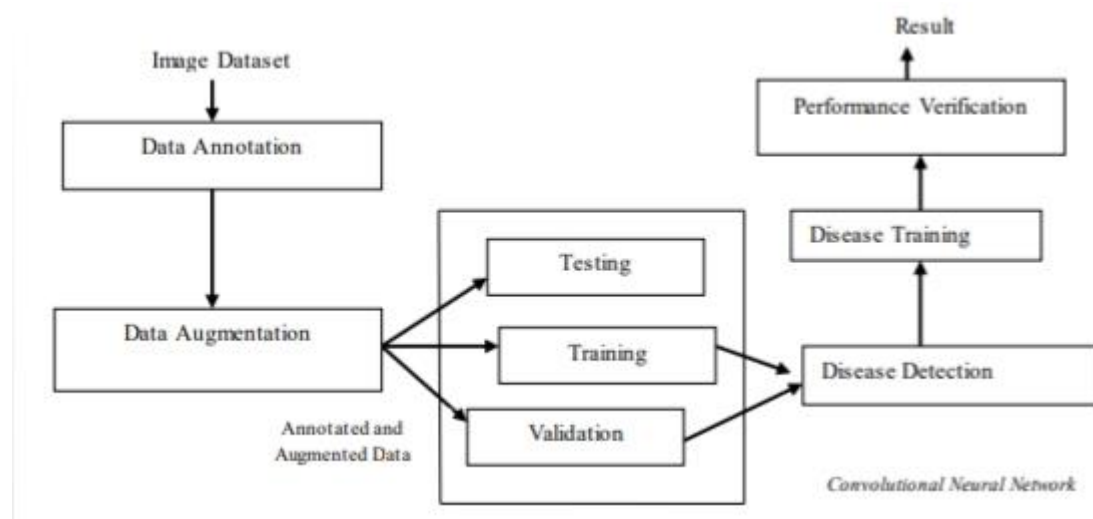
Reference 2 :- Image Analysis and Detection of Tea Leaf Disease using Deep Learning

Introduction :- The widespread development of deep learning has directed to the increasing research interest in image recognition technologies, which enables in the field of computerized image categorization and finding of plant diseases. India is the world's largest tea consumer and the second largest tea producer. Diseases can harm the plant's health, which in turn affects its growth. There are a number of tea leaf diseases that target the rate at which the crop grows. The purpose of this paper is to develop a deep CNNs called LeNet to discover the tea plant diseases from leaf image set. As a result, LeNet the perfect CNN model used in further applications to improve the diagnostic measurement of tea leaves as well as other plant leaves.

Dataset used :- The data leaves contain 80 images of selective tea leaf's diseases, named as blister blight, red scab, red leaf spot and leaf blight. The image is collected from plant village dataset.

Methods Used :-

This paper presents methods of image analysis through edge-based thresholding; K- means clustering and region-based extraction. The extracted regions of objects were compared to the mean difference value between the object. The correlation of objects is calculated in two images, to measure the similarity index between the images. The PCA algorithm is used to reduce the quantity of the feature with different orientations to improve the classification accuracy. This approach tells us that the Kmean clustering approach is good in yielding high accuracy compare with other approaches.



Results :-

Tea Leaf blight	Blight disease	Tea Red Leaf spot	Tea Red Scab	Sensitivity	Accuracy	MCA
Tea Leaf blight	90	1	0	86.49%		
Blight disease	1	117	0	98.32%	90.23%	90.16%
Tea Red Leaf spot	0	0	95	85.59%		
Tea Red Scab	0	3	0	94.87%		

Remarks :- The computational complexity required for using conventional neural network is significantly reduced using this model. The high adaptation to internal failure of CNNs permits the utilization of incomplete or uncertain foundation pictures, subsequently adequately improving the precision of picture acknowledgment. The consequences of the current examination underline the chance of utilizing CNNs in the recognizable proof and assessment of tea plant sicknesses, which could altogether improve malady acknowledgment for tea ranch horticulture. In spite of the fact that the exactness of LeNet's sickness characterization isn't 100%, upgrades to the present strategy might be actualized in future examinations. In our future work, the LeNet model has been utilized to distinguish other yield ailments, for example, tomato leaves and mulberry leaves. However, this should be further investigated according to the specific circumstances of the disease.

Reference 3 :- TeaViTNet: Tea Disease and Pest Detection Model Based on Fused Multiscale Attention

Introduction :- The tea industry, as one of the most globally important agricultural products, is characterized by pests and diseases that pose a serious threat to yield and quality. These diseases and pests often present different scales and morphologies, and some pest and disease target sizes can be tiny and difficult to detect. To solve these problems, we propose TeaViTNet, a multi-scale attention-based tea pest and disease detection model that combines CNNs and Transformers. First, MobileViT is used as the feature extraction

backbone network. MobileViT captures and analyzes the tiny pest and disease features in the image via a self-attention mechanism and global feature extraction. Second, the EMA-PANet network is introduced to optimize the model's learning and attention to the *Apolygus lucorum* and leaf blight regions via an efficient multi-scale attention module with cross-space learning, which improves the model's ability to understand multi-scale information. In addition, RFBNet is embedded in the module to further expand the perceptual range and effectively capture the information of tiny features in tea leaf images. Finally, the ODCSPLayer convolutional block is introduced, aiming to focus on acquiring richer gradient flow information. The experimental results show that the TeaViTNet model proposed in this paper has an average accuracy of 89.1%, which is a significant improvement over the baseline network MobileViT and is capable of accurately detecting *Apolygus lucorum* and leaf blight of different scales and complexities

Datasets Used :- In this study, we conducted detailed field research on Maoshan Tea Factory in Jurong City, Jiangsu Province, China, in which the staff of the tea factory pointed out that leaf blight and *Apolygus lucorum* are two common challenges in tea growth. Of note, leaf blight is a disease, while *Apolygus lucorum* is an insect pest. These hazards have significant adverse effects on the quality of tea, so this paper takes these two pests and diseases as the research object. In this study, we used both a DJI Mavic 3T drone and a cell phone to collect images of tea pests and diseases. The drone collected images of tea trees at a height of 2 m to ensure the clarity and detail of the images. In this way, we collected drone images with a resolution of 1920×1080 , while we also used a smartphone for image collection from a ground perspective. The smartphone has an image resolution of 3024×4032 , which provides a higher resolution image that helps to capture more detailed pest and disease characteristics. By combining these two approaches, we screened 450 high-quality images Agronomy 2024, 14, x FOR PEER REVIEW of tea pests and diseases. These images cover pest and disease conditions from different 5 of 24 viewpoints from the air and ground, providing a rich and diverse dataset for our study.



Data Augmentation :- The sample size of the dataset obtained above is small, which is not enough to support the effective training and evaluation of deep learning models. To solve the problem of a small sample size, this paper uses three image enhancement methods based on random rotation, brightness adjustment, and adding noise. Random rotation can increase the sample diversity, simulate the same sample taken from different angles, and allow the model to learn that the features of pests and diseases can still be recognized even from different viewpoints. Brightness adjustments can simulate the actual lighting variations, improve the accuracy of the model in real applications, and enhance the feature saliency at the same time. Adding noise improves the robustness of the model, making it more resistant to noise and less dependent on clean data. These enhancement methods aim to generate more diverse image samples, thus improving the robustness and accuracy of the model in different scenes.

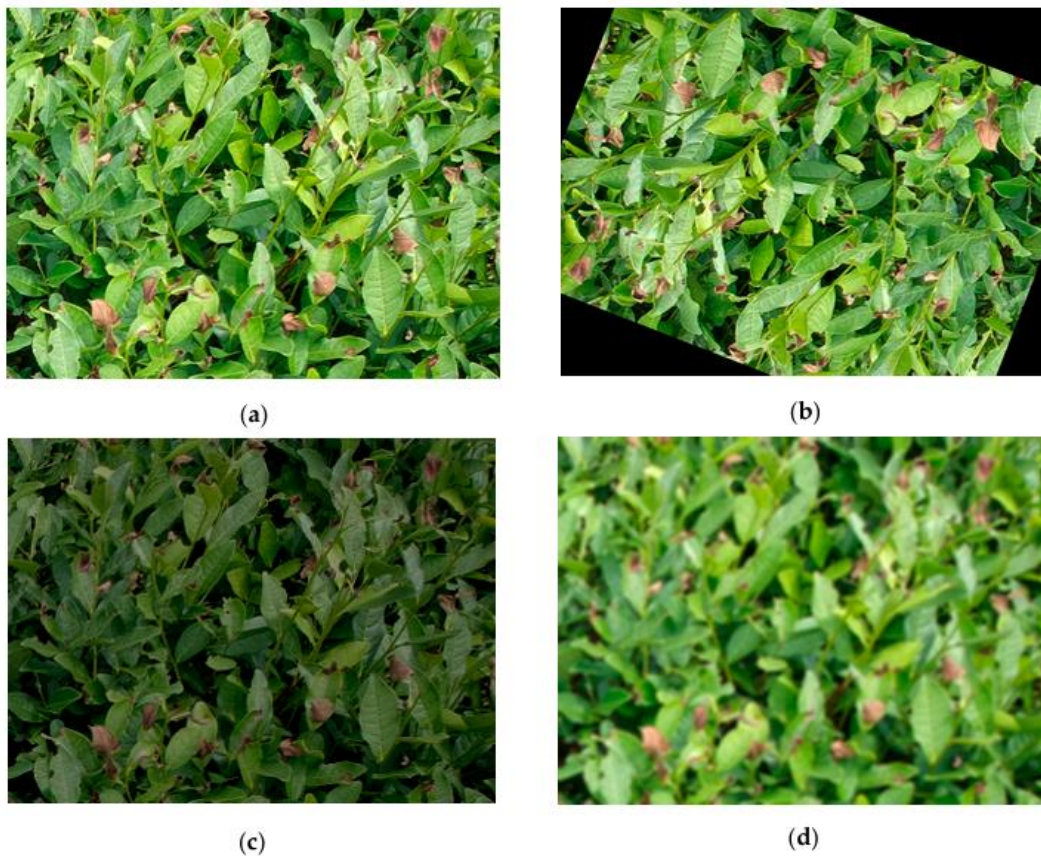


Figure 2. Sample data augmentation for tea pests and diseases: (a) original image; (b) random rotation; (c) brightness adjustment; (d) adding noise.

Distribution	Before Data Augmentation	After Data Augmentation
Images	450	2100
<i>Apolygus lucorum</i>	1583	9629
Leaf blight	1962	13,260

Distribution	Training	Validation
Images	1680	420
<i>Apolygus lucorum</i>	7703	1926
Leaf blight	10,608	2652

Methods used :-

In this paper, we propose a detection model, TeaVitNet, which combines CNNs with Transformers. The model network structure consists of 7 of 24 an input, backbone, neck, and head

Backbone: MobileViT is used as the feature extraction network of the model; the network module inherits the advantages of lightweight CNNs with fewer numbers of parameter. Of parameters and less computation and adopts a visual Transformer based on the mechanism of multi-head self-attention to encode the global information to construct a light weight feature extraction network , which can sufficiently extract the information of the local and global samples , and generate the feature maps of different scales to the neck network.

Neck: Multi scale attention EMA-PANet is used as a neck feature fusion network; EMA-PANet has three feature fusion branches and a fusion feature enhancement module . An efficient multi-scale attention mechanism is added to the output nodes of the network to effectively capture cross-dimensional interactions and establish inter-dimensional dependencies, and the fusion of multi-scale contextual information makes the network pay more attention to useful information. Its network structure mainly consists of RFBNet , ODCSPLayer , and EMA.

Head: The feature maps generated by the neck are used to predict target bounding boxes , category probabilities , and confidence levels for the bounding boxes . Multiple bounding boxes are predicted for each anchor location , while the network also predicts the category probability and object confidence of the targets contained within each bounding box.

Input Layer	Output Size	Stride	Repeat
Image	640×640		
Conv	320×320	2	1
MV2	320×320	1	1
MV2	160×160	2	1
MV2	160×160	1	2
MV2	80×80	2	1
MobileViT Block	80×80	1	1
MV2	40×40	2	1
MobileViT Block	40×40	1	1
MV2	20×20	2	1
MobileViT Block	20×20	1	1

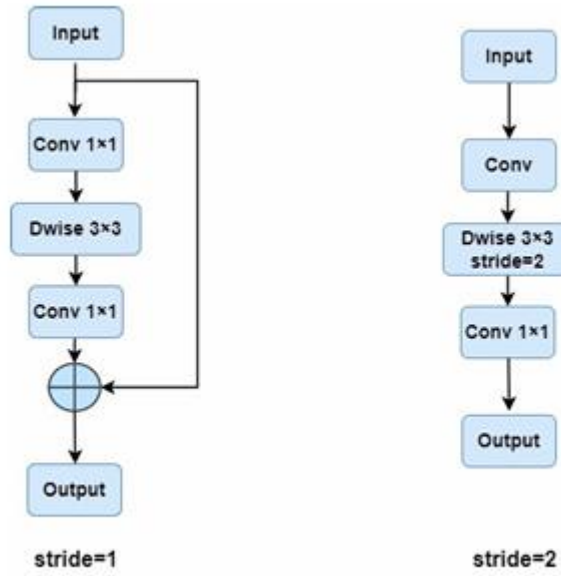


Figure 4. MV2 block.

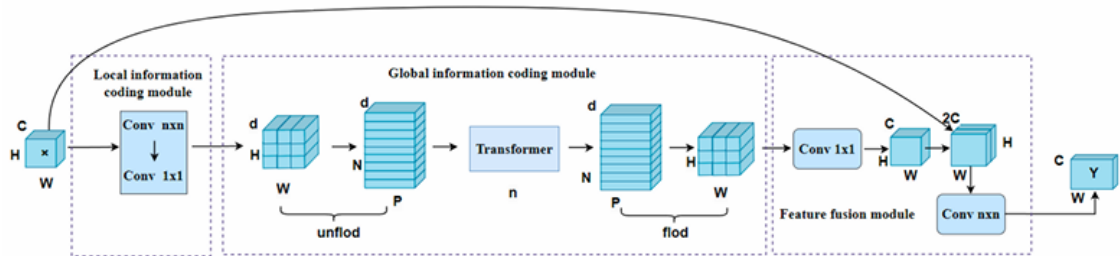


Figure 5. MobileViT Block.

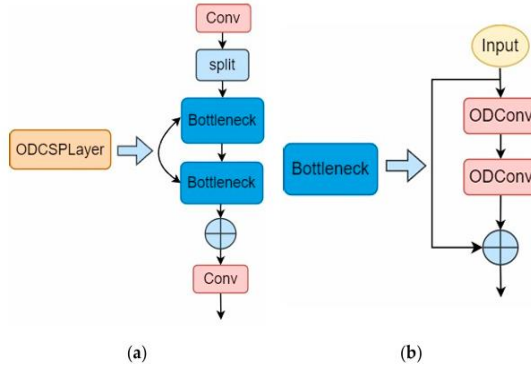


Figure 6. (a) ODCSPLayer module; (b) bottleneck.

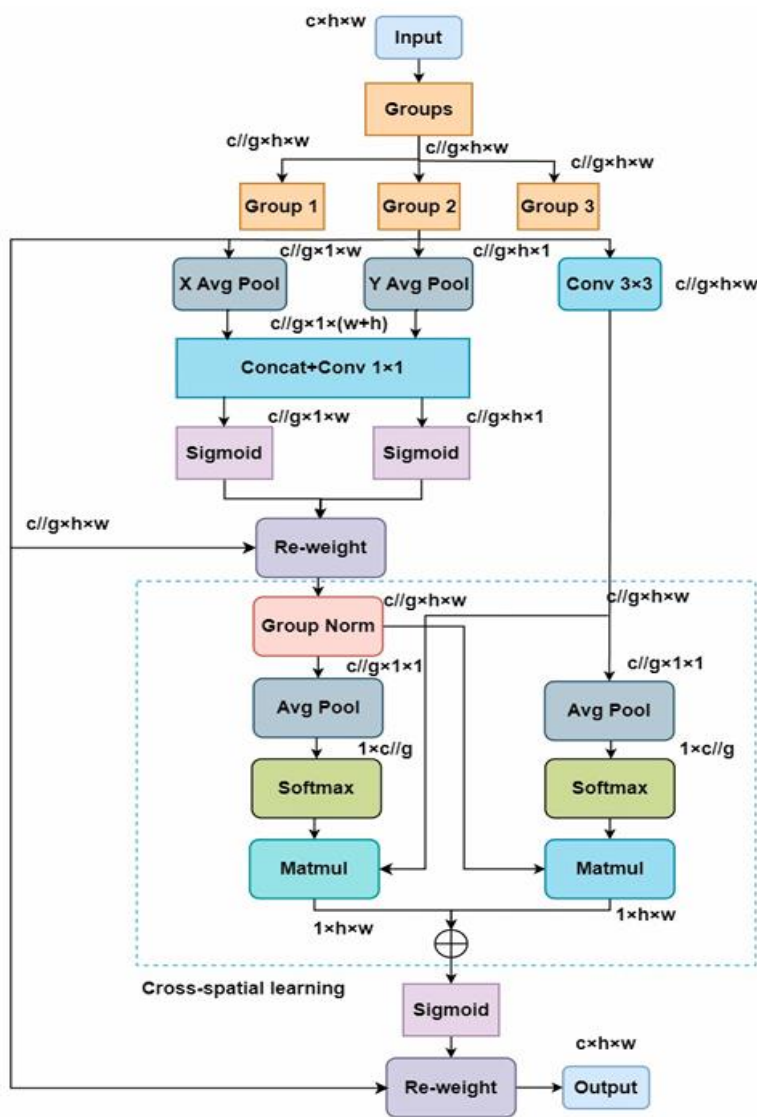


Figure 7. Diagram of EMA module.

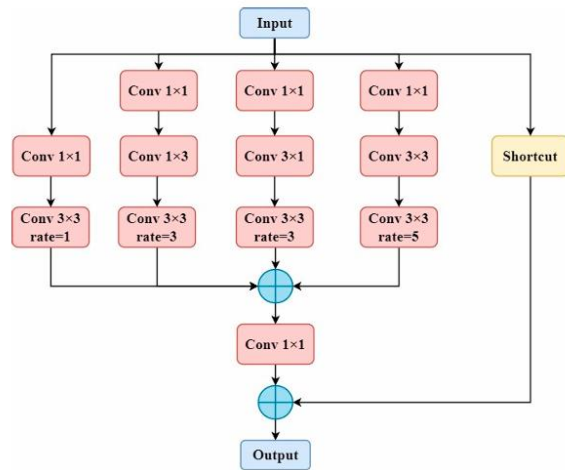
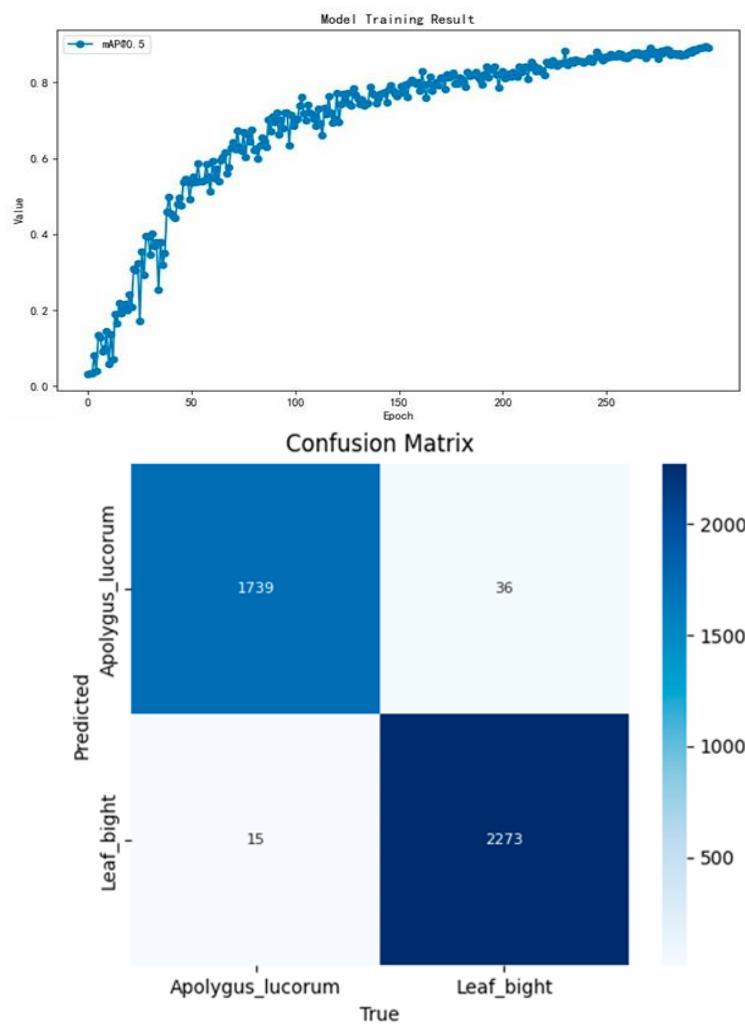


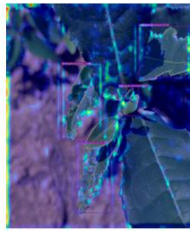
Figure 8. Structure of RFBNet.

Results : -

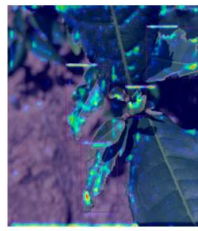




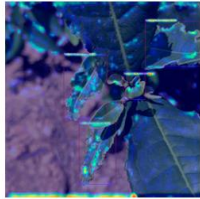
Input image



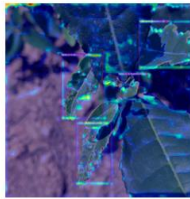
Baseline



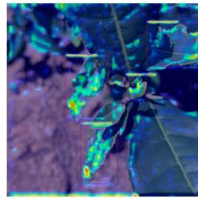
CBAM



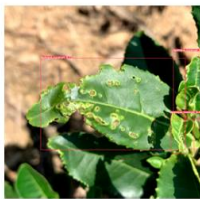
ECA



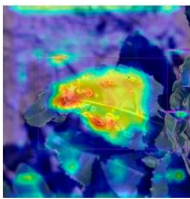
SE



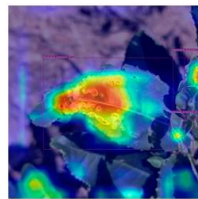
EMA



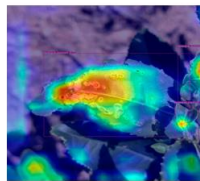
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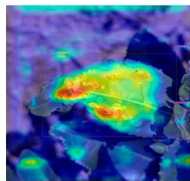
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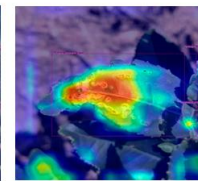
CBAM



ECA



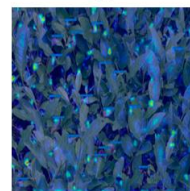
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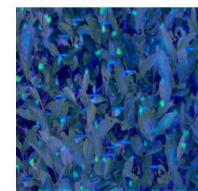
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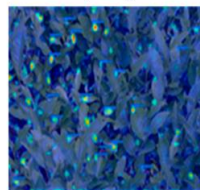
Input image



Baseline



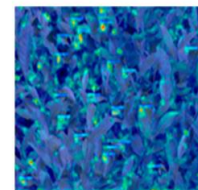
CBAM



ECA



SE



EMA



Remarks :- In the future, the TeaViTNet model can continue to be refined and optimized. For example, the TeaViTNet model pre-trained in the identification of tea pests and diseases can be used for migration learning to identify pests and diseases of other crops, thus promoting the development and progress of the agricultural field. Meanwhile, real-world application tests are conducted to deploy the model into real agricultural scenarios and conduct field tests to verify its stability and reliability in complex environments. Overall, this study provides a valuable direction for exploration in the field of tea pests and disease identification. Future research will not only continue to improve the model but also expand its application scope, provide technical support for intelligent monitoring of agricultural pests and diseases, and promote the development of agricultural automation and intelligence